

BBA 2421

Business Mathematics



UNIT 1

Basic Algebra and Mathematics of Buying



Learning objectives

- By the end of this unit, students should be able to
 - Understand algebraic functions
 - Compute arithmetic and geometric progressions
 - Compute and interpret discounts
 - Compute and interpret mark up
 - Compute and interpret mark downs



Basic Algebra: Algebraic functions



What is Algebra?

- Algebra refers to a set of rules for manipulation of systems of linear and nonlinear relationships.
- It is sometimes referred to as vector and matrix algebra.
- Relationships do occur in a system because they contain variables that are in common.
- Note:
 - **Reading assignment on algebraic functions and presentations will be made in the next class.**



Arithmetic and geometric progressions

Arithmetic Progression

- Consider these two common sequences

1, 3, 5, 7, ...

and

0, 10, 20, 30, 40, ...

- These sequences each start with a particular first term, and then to get successive terms we just add a fixed value to the previous term.

- The difference between consecutive terms in each sequence is a constant.
- Let a = first term of an AP and d = common difference, then the n th term of a sequence will be given by:

$$a + (n - 1)d$$

- Note:**
 - The remainder on AP is reading assignment and presentations will be made in the next class.



Arithmetic and geometric progressions

Geometric progression

- Consider the sequence
 $2, 6, 18, 54, \dots$
- Each term in the sequence is 3 times the previous term. And in the sequence
 $1, -2, 4, -8, \dots,$
- Each term is -2 times the previous term.
- Sequences such as these are called **geometric progressions, or GPs** for short.

- Let a =first term and r =common ratio then in general geometric progression can be expressed as:

$$ar^0, ar, ar^2, ar^3, \dots$$

- Hence the n th term could be computed by:

$$ar^{n-1}$$

- **Note:**

- The remainder on AP is reading assignment and presentations will be made in the next class.



Mathematics of Buying

- Retail is a huge business that provides millions of jobs around the world.
- Retailers make a profit by purchasing items at one price and then selling them for a higher price.

- A **supply chain** is made up of:

Raw materials suppliers $\Rightarrow$ **manufacturers** $\Rightarrow$ **wholesalers** (or distributors)

$\Rightarrow$ **retailers** $\Rightarrow$ **the consumer**



Invoices and Trade Discounts

- An **invoice** is a document issued by a seller to a buyer as a record of a transaction.
- It shows the seller, buyer, items purchased, quantity, price, and any discounts. That is, it shows the shipping and insurance charges, and the invoice total, which is the sum of the extension totals.
- Note that trade and cash discounts are never applied to shipping and insurance charges.
- The following are the common shipping term that appear on invoices: **free on board (FOB) shipping point, free on board (FOB) destination** and **free alongside ship**.



- **FOB Shipping Point**

- Under such an arrangement, the buyer pays for shipping costs and ownership of goods is transferred to the buyer prior to the shipment.

- **FOB Destination**

- Under such an arrangement, the seller incurs the shipping charges and retains ownership until the goods reach the destination.

- **Free alongside ship (FAS)**

- Goods are delivered to the dock with all freight charges to that point paid by the seller.

- **Note:**

- Since goods can be damaged, lost, or stolen during shipment, it is vital for everyone involved in the transaction to know which firms own products that are in shipment.
 - This is the only way it will be clear who will incur the liability or pay for the goods lost or damaged while in shipment.



Trade discount

- Offered to businesses or individuals who buy an item that is to be sold or used to produce an item that will then be sold.
- The seller usually gives the price of an item as its **List price** (i.e. the suggested price at which the item can be sold to the public).
- The seller then gives a trade discount that is subtracted from the list price. Therefore, the amount paid by the buyer is:

$$\text{Net cost/net price} = \text{list price} - \text{trade discount}$$



- **Net cost** (cost of an item to the business) and **net price** both refer to the amount paid by the buyer.

Example 1 : Calculating a single trade discount

Assume Game stores is given a list price and trade discount of K100 and 25% respectively. What is the net cost?

$$\text{Trade discount} = \frac{25}{100} \times K100 = K25$$

- Therefore, net cost = $K100 - K25 = K75$.
- In Example 1 a single discount of 25% was offered.
- In some cases, two or more discounts are combined into a series or chain discount.



- A series discount simply refers to more than one discount in a series. For instance, the series discount 20/10 implies:
 - i. First apply the 20% discount and obtain the initial net cost. Thereafter,
 - ii. Apply the 10% trade discount to the initial net cost in (i) above to obtain the final net cost

Example 2

- Let us assume that Game stores was offered a series discount of 20/10 on kitchen utensils with a list price of k100, 000. Find the net cost after the series discount.



Solution

- Step 1: Find initial net cost after the first discount
 - Apply the 20% discount
 - $\text{Discount} = 20\% \times K100 = K20$
 - Initial Net Cost/Amount after first discount $= K100 - K20 = K80$
- Step 2: Find the final net cost by using the second discount
 - Apply the 10% to the initial net cost in step 1 i.e $K80$
 - $\text{Discount} = 10\% \times K80 = K8$
 - Final Net Cost $= K80 - K8 = K72$
- Note:
 - Never add series discounts together e.g. the series discount 20/10 is not the same as 30% discount. Instead, apply the discounts sequentially, or one after the other.



- We can also compute series discounts by using ***complements***.
- The **complement of the discount** is the portion paid. e.g a 10% trade discount means that 90% is paid by the buyer. Thus, 90% is the complement of the discount
- To find the complement of a discount, simply subtract the decimal equivalent of each single discount from 1.
- The complement of a discount can be used to find the **net cost equivalent**, which is then used to find the **net cost** after the discount.

Example 3

Refer to example 2 above. Find the discount using net cost equivalent method?



Solution

- **Step 1:** Find the complement of each discount in the series discount.

Discount	Decimal form	Complements
20%	0.2	$1-0.2=0.8$
10%	0.1	$1-0.1=0.9$

- **Step 2:** find the net cost equivalent by multiplying the complements from step 1.
 - Net cost equivalent= $0.8 \times 0.9 = 0.72$
- Note:
 - Do not round off the net cost equivalent because doing so will often lead to an inaccurate net cost.



- **Step 3:** Find the net cost after the discount:
 - The net cost after the discount is the list price times the net cost equivalent.
 - Net cost after discount = List price $\times$ Net cost equivalent

$$= K100 \times 0.72 = K72$$
- To find the actual discount, subtract the net cost from the original price
 - **Discount = Original price - Net cost after discount**

$$= K100 - K72 = K28$$
- From the foregoing, **the net cost equivalent or percent paid** is the product of the complements. That is:
 - Net cost equivalent = First complement $\times$ Second complement $\times \dots$
 - Net cost = List price $\times$ Net cost equivalent



- Trade discount do change because:
 1. Size of order placed with suppliers
 2. Seasonal fluctuations
 3. Geographic location
 4. Level of competition
 5. Price changes



Cash Discounts: Ordinary Dating Methods

- Class objectives:
 - 1) Calculate net cost after cash discount
 - 2) Use the ordinary dating method
 - 3) Determine whether cash discounts are earned
 - 4) Postdated invoices and calculating cash discounts.



- **Cash discounts** are offered by sellers to encourage prompt payment by customers.
- Prompt payment by customers reduce the seller's appetite to borrow money when new investment opportunities become available.
- To compute the net cost, subtract any trade and cash discounts from the list price.

$$\textbf{Net cost} = (\textbf{List Price} - \textbf{Trade discount}) - \textbf{Cash discount}$$

- Note
 - If an invoice amount includes shipping and insurance charges, subtract these charges first, before a cash discount is taken. Then add them back to find net cost after the cash discount is subtracted.



- There are a number of methods used to calculate cash discounts, though most of these methods are based on the ordinary dating methods (ODM).
- Ordinary Dating Method is one of the two most common methods used today.
- Most companies state the exact cash discount on the invoice, and the ordinary dating method of cash discount is expressed as
$$2/10, n/30 \text{ or } 2/10, \text{net } 30$$
- And is read as: “two ten, net thirty”.



- **Interpretation**

- A 2% discount applies if the invoice is paid within 10 days of the invoice date.
 - If no payment is made within 10 days, then the full invoice amount must be paid within 30 days of the invoice date.
 - And if no payment is made by the end of 30 days from the invoice date, then a penalty may apply.
- Note: Use the number of days in each month to find the due date of an invoice. Use the knuckle method to determine the number of days in a month.

Number of days in a month	
30-day months	April, June, September, November
31-day months	January, March, May, July, August, October, December
Exception	February (28, 29)



- **Case 1: Determining When Cash Discounts Are Earned**

- Find the date that an invoice is due by counting from the next day after the date of the invoice. (i.e. the date of the invoice is never counted) OR
- To determine the due date:
 - Add the given number of days to the invoice date. e.g to determine 10 days from April 7th, add the given number of days to the invoice or start date. $7+10=17$ i.e. the due date or 10 days from April is April, 17.

- **Case 2: When The Due Date /Net Payment Date Falls In The Following Month**

- To determine the due date, the following steps will be followed:
 1. Determine the number of days remaining in the current month by subtracting the invoice date from the total number of days in that month. Then



- 2. Find the number of days in the next month needed to equal the discount period or net payment period. For example, to determine the due date that is 15 days from October 20th, we proceed as follows:
 - Step 1:
$$\begin{array}{r} 31 \text{ total number of days in October} \\ -20 \text{ the start date (i.e. October 20th)} \\ \hline =11 \text{ days remaining in October} \end{array}$$
 - Step 2:
 - But 15 days is the given number of days.
 - Thus, to find the number of days needed to equal the discount period or net payment period, subtract the 11 days remaining in October from the 15 days i.e. $15 - 11 = 4\text{th November}$.
 - Therefore, 4th November is 15 days from October, 20th.



Example

- A Game store invoice is dated January 2 and offers terms of 2/10, net 30.
- a) Find the last date on which the 2% discount maybe taken

Answer: Since the due date falls in the same month, add the given number of days to the start/invoice date. That is, $2 + 10 = 12$. Thus, the last date is 12th January.

- b) The net payment date.

Answer: The net payment date is:

31 total number of days in January
-2 the start date (i.e. January 2nd)
=29 days remaining in January



- To find the number of days needed to equal the discount period or net payment period, subtract the 29 days remaining in January from the given 30 days.
- That is, $30 - 29 = 1\text{st February}$. Therefore, 1st February is 30 days from 2nd January.

In-class Exercise

An invoice is dated June 8 and offers terms of 3/15, net 30.

Find

- i. The last date on which the 3% discount may be taken, and
- ii. The net payment date



Finding The Amount Due On The Invoice

Example

Assume an invoice received by Game store for K840 is dated July 1 and offers terms of 2/10, n/30. If the invoice is paid on July 8 and the shipping and insurance charges, which were FOB shipping point, are K35.60. Find the total amount due.



Solution

- Invoice paid in 7 days (i.e. $8-1=7$) which is less than the 10 day requirement identified by the terms 2/10. Thus, the 2% cash discount applies.

$$\begin{aligned}\text{Cash discount} &= 2\% \text{ of } 840 \\ &= 16.80\end{aligned}$$

$$\begin{aligned}\text{Initial amount due} &= \text{invoice amount} - \text{cash discount} \\ &= 840 - 16.80 \\ &= 823.20.\end{aligned}$$

- However, the shipping and insurance chargers must be added to the amount.

$$\begin{aligned}\text{Total to be paid} &= \text{amount due} + \text{shipping and insurance charges} \\ &= 823.20 + 35.60\end{aligned}$$

$$\text{Total Amount} = 858.80$$

Note: Shipping and insurance charges are not eligible for cash discounts. First, deduct applicable trade and cash discounts. And then add the shipping and the insurance charges to find total amount due.



Calculating cash discounts when an invoice is postdated

- Recall that under the ordinary dating method, the cash discount date and net payment date are both counted from the date of the invoice.
- In some cases, the invoice is post-dated, and this may be done to give the buyer more to take advantage of the cash discount on the invoice or may be done to fit the accounting practices of the seller.
- The seller will place a date that is after the actual invoice date, sometimes labelling AS OF. For instance

Date 07/25	AS OF 08/01
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- Both the cash discount period and the net payment date are counted from 08/01. This gives the purchaser additional time to pay the invoice and receive the discount.



Example

- Assume an invoice for Game stores for shipment of kitchen utensils from South Africa is dated October 21 AS OF November 1 with terms 3/15, n/30

i. Find the last date on which the cash discount maybe taken, and

Answer: Start with the postdate (AS OF) of November 1:

Therefore, the last date for taking the discount is

$$\begin{array}{r} 1 \\ + \quad 15 \\ \hline =16^{\text{th}} \text{ November} \\ \hline \end{array}$$

b) The net payment date



- The net payment date is
 - 30 days in November
 - 1 start date on invoice
 - =29 days remaining in November.
- But 30 Total number of days on the terms
 - 29 Days remaining in November
 - =1st December as the Net payment date.
- Therefore, the net payment date is December 1.



In-class exercise

An invoice for Game stores from South Africa is dated August 24 AS OF September 1 with terms of 1/10, n/30

- i. Find the last date on which the cash discount maybe taken
- ii. The net payment date.

Example

An invoice is dated May 18 and offers terms of 4/10, 3/25, 1/40, n/60

1. Find the three final dates for each cash discount, and
2. The net payment date



Solution

- a) The 3 final cash discounts are
 1. 4% if paid by May 28 (10 days from May 18)
 2. 3% if paid by June 12 (25 days from May 18)
 3. 1% if paid by June 27 (40 days from May 18)
- b) The net payment date is July 17 (20 days beyond the cash discount period).



Other dating methods for cash discounts

- **End of month:** Dating denoted by e.g. 3/10 EOM OR 3/10 prox.
 - The above entails that 3% may be taken as cash discount if payment is made by the 10th of the month that follows the sale
 - E.g. An invoice dated July 14 with terms of 3/10 EOM would have a discount date 10 days from the end of the month, i.e. 10th of August.
- **Receipt of goods dating:** Denoted by e.g. 3/15 ROG
 - This method is used when the shipping time is long, and offers cash discounts determined from the date on which goods are actually received.
 - Therefore, 3/15 ROG allows 3% cash discount if the invoice is paid within 15 days from receipt of goods.
 - E.g. if the invoice is dated March 5 and the goods were received on April 7, then the last date to take 3% cash discount would be April 22 ($7+15=22$)
 - The next payment date, since it's not stated, is 20 days after the last discount date, or May 12th (April 22 + 20 days)



- **Extra dating** (extra, ex., or x)
 - This gives the buyer additional time to take advantage of a cash discount, and the discount is expressed as, e.g.;
 - 2/10-50 extra or
 - 2/10-50 ex. Or
 - 2/10-50 x
- The above expressions allow a 2% cash discount if the invoice is paid within $10+50=60$ days from the date of the invoice.
- The discount is denoted as 2/10-50 ex. (rather than 2/60) to show that the 50 days are extra or in addition to the 10 normal days offered.



Mathematics of Selling

- The success of a business depends on a number of factors.
- These include, among others, controlling costs, hiring competent staff as well as the price at which the product is sold.



- The following terms are used in selling
 - i. **Cost:** The amount paid to the seller after trade & cash discounts have been taken.
 - ii. **Selling price:** The price at which the products are sold to the public
 - iii. **Mark-up, margin or gross profit:** This is the difference between the selling price and cost. Note: the above terms are often used interchangeably.
 - iv. **Operating expenses or overhead:** These include expenses of operating the business such as rent, wages, salaries, utilities, insurance and advertising.
 - v. **Net profit (net earnings):** The amount remaining for a business after operating expenses and the cost of goods and services have been paid. Note that income tax is computed on net profit.



Basic Formula for Mark-up

- Before offering the product for sale to the general public, sellers markup the cost of the product. Thus

$$\textit{Selling Price} = \textit{Cost} + \textit{Markup}$$

$$S = C + M$$

- The markup that the seller sets should be large enough to cover all operating expenses and have enough left over for profit.
- Note that sometimes, the markup maybe small to encourage customers to buy or when products go on sale
- Recall that cost is the amount paid to the seller and that any markup left over after paying operating expenses is profit.



Markup On Cost Or Markup Based On Cost

- This is the markup which is stated as a percentage of cost. Thus the phrase markup on cost is used and its given by:

$$\text{Markup on Cost} = \frac{\text{Amount of Markup}}{\text{Cost}} \times 100$$

Example 1

- Suppose a book costs K100 and sells for K130. Find the markup on cost.

Answer: $\text{Markup} = \text{Selling Price} - \text{Cost} = 130 - 100 = 30$

- Thus the markup on cost is:

$$\text{Markup on Cost} = \frac{30}{100} \times 100 = 30\%$$



Example 2

- Suppose the markup on a book is set at K16, which is 50% based on cost. Find
 - a. The cost
 - b. Selling price

Solutions

a. $\text{Markup} = 50\% \times \text{Cost}$

$$16 = 0.5 \times C$$

$$C = K32$$

b. Recall that

$$S = \text{Markup} + \text{Cost} = 16 + 32 = 48$$



- Noted that:

- i. Cost as percentage of cost is 100%
- ii. Selling price as a percentage of cost is 150%
- iii. Markup as a percent of cost is 50%

$C\% + M\% = SP\%$ This is true for any identity

$$100\% + 50\% = 150\%$$

Example 3

Mulenga best sells his bed space for K300. If the markup on cost is 40%, find the amount that Mulenga best pays for the room and the amount of markup.



Solution

- We know that;
 - i. Cost expressed in terms of itself is 100% and Markup on cost is 40%
 - ii. Selling Price is 140% of Cost
 - iii. However, we don't know the base or absolute cost amount
 - iv. We use the basic % formula and make cost the subject of the formula

$$\begin{aligned}\frac{S}{Cost} \times 100 &= 140\% \\ S &= 1.4 \times Cost \\ Cost &= \frac{Selling\ price}{1.4} = \frac{300}{1.4} = 214.2857\end{aligned}$$

- The amount of markup is $M = S - C = 300 - 214.2857 = K85.7143$



100%	C	K214.2857	Base
40%	M	K85.7143	
140%	S	K300	

Example 4

- Suppose Nera sells laptops and the price of a laptop is K549.99. Besides this, Nera has operating expenses of 29.5% and wants a 5.5% profit both based on cost, on this item. Find the cost and markup on cost.

Solution

- First find the total markup on cost: Add operating expenses % of cost and profit % of cost to find the total % markup on cost.



- $M\% \text{ of cost} = \text{Operating Expenses \%} + \text{Profit \%} = 29.5\% + 5.5\% = \mathbf{35\%}$
- Now we know

100%	C	?
<u>35%</u>	<u>M</u>	<u>?</u>
?%	S	549.99

- The selling as a % of cost is $100\% + 35\% = 135\%$
- The cost (base) is $\text{Cost} = 549.99 / 1.35 = \mathbf{407.40}$
- The markup is; $\mathbf{M = S - C} = 549.99 - 407.40 = \mathbf{142.59}$



Markup on selling price

- Sometimes sellers compare business operations to sales revenues and thus often prefer to use markup on selling price.
- In this scenario, the markup on selling price is stated as a % of the selling price.
- The formula for markup on selling is given by
- Markup on Selling Price = $\frac{\text{Amount of Markup}}{\text{Selling Price}} \times 100$
- From above, we note that the selling price becomes the base when solving problems about the markup on selling the price.



Example

- Assume a product is sold for K39.99 and the cost of each pair is K35. Calculate the markup on selling price.

Solution

- Set up the problem as;

?%	C	K35
<u>?%</u>	<u>M</u>	<u>K?</u>
100%	S	K39.99

- $M = S - C = 39.99 - 35 = 4.99$**
- Using the basic % formula to find the markup on selling price: $\frac{4.99}{39.99} \times 100 = 12.5\%$



- **Determining the % markup on cost & equivalent markup on selling price.**
 - At times, a seller may wish to compare the markup on cost with the markup on selling price. For example, a salesperson selling to both manufacturers using markup on cost and retailers using markup on selling price might want to make conversions from one markup to the other for comparison purposes.

Example

- Suppose a product cost K2.10 and is sold for K3.2. Find the percentage markup on cost as well as the percentage markup on selling price, and interpret your findings.



Solution

- Set up the problem using cost as the base.

100%	C	K2.1	base
<u>?%</u>	<u>M</u>	<u>?</u>	
?%	S	K3.20	

- $M = S - C = 3.2 - 2.1 = \text{K1.10}$
- Markup as a percentage of the cost is: $(1.10 / 2.1) \times 100 = 52.4\%$



- To find markup as a percentage of the selling price, set up the problem using selling price as the base.

?%	C	K2.1	
<u>?%</u>	<u>M</u>	<u>K1.10</u>	
100%	S	K3.2	Base

- Thus, markup as a percentage of selling price is: **$(1.10 / 3.20) \times 100 = 34.4\%$**
- **Implications:** This means that a 52.4% markup on cost results in the same kwacha markup as a 34.4% markup on selling price OR 52.4% markup on cost is equivalent to a 34.45 markup on selling price.



- **Converting markup percentage on cost to markup on selling price.**

- We can also use conversion formulas to make markup comparisons.

$$\%Markup \text{ on Selling Price} = \frac{\%Markup \text{ on Cost}}{100\% + \text{Markup on Cost}}$$

Example

- Convert a markup of 25% on cost to its equivalent markup on selling price.

Solution

- Using the formula

$$\begin{aligned}\%Markup \text{ on Selling Price} &= \frac{\%Markup \text{ on Cost}}{100\% + \text{Markup on Cost}} \\ &= \frac{25\%}{100\% + 25\%} = \frac{25\%}{125\%} = 20\%\end{aligned}$$



- **Converting percentage markup on selling price to markup on cost.**

- We can also use conversion formulas to make markup comparisons.

$$\%Markup \text{ on cost} = \frac{\%Markup \text{ on selling price}}{100\% - Markup \text{ on selling price}}$$

Example

- Convert a markup of 20% on cost to its equivalent markup on cost.

Solution

- Using the formula

$$\begin{aligned}\%Markup \text{ on cost} &= \frac{\%Markup \text{ on selling price}}{100\% - Markup \text{ on selling price}} \\ &= \frac{20\%}{100\% - 20\%} = \frac{20\%}{80\%} = 25\%\end{aligned}$$



Markdown

- It is imperative that slow moving inventories are moved out of the store to make room for items that are in demand.
- One way to move stocks/inventories out of the store is to mark them down
- Thus, Markdowns are used to stimulate sales volume.
- Definition:
 - Markdown is the difference between the original selling price and the reduced selling price. In other words, markdown is simply just the slashing of prices in order to stimulate sales.



- The selling price after the markdown is called:
 - i. Reduced price
 - ii. Sale price
 - iii. Actual selling price
- The formula for markdown is given by:

$$\text{Reduced Price} = \text{Original Price} - \text{Markdown}$$

Example

- Assume game stores marks-down its stoves. Find the reduced price if the original price was K2879 and the markdown is 30% of the original price.



Solution

- Markdown: 30% of the original price

$$0.3 \times 2879 = 863.70$$

- The reduced price.

$$\begin{aligned} \text{Reduced price} &= \text{original price} - \text{markdown} \\ &= 2879 - 863.70 = 2015.30 \end{aligned}$$

- 70% of the original price.
- **How to find markdown as a percentage of the original price**
 - This is also referred to as the rate and is given by the following formula

$$\text{Rate} = \frac{\text{Markdown}}{\text{Original price (base)}}$$



Example

- If the retail value of books is K785 and the books are later sold at a reduced price of K530, What is the % of markdown on the original price.

Solution

$$\text{Reduced Price} = \text{Original Price} - \text{Markdown}$$

$$\text{Markdown} = 785 - 530 = 255$$

- Thus, the Rate = $(255 / 785) * 100 = 0.3248 * 100 = 32\%$

Example

- Assume Game stores offers a laptop bag at a reduced price of K63 after a 25% markdown from the original price. Find the original price.
- **Answer:** After 25% markdown, the reduced price of K63 represents 75% of the original price.

$$75\% \times \text{Original Price} = \text{Reduced price}$$

- Original price therefore = $63 / 0.75 = K84$



- **Definition of terms associated with loss**

- We know that sellers always want to maximise profits by selling the item at the full price.
- Unfortunately, sellers have to markdown prices of slow moving items to create space for new items.
- But marking down prices reduces the profits and sometimes results in a loss depending on the amount of markdown.



- The following are terms are worth noting:

1. Break-even point

- This is the selling price that covers just the cost of the item plus the associated overhead, including operating expenses.
 - In this case, the firm neither makes a loss nor profit at the break-even point

2. Reduced net profit

- Occurs when the item is marked down from the original price but is still sold for more than the break-even point

3. Operating loss

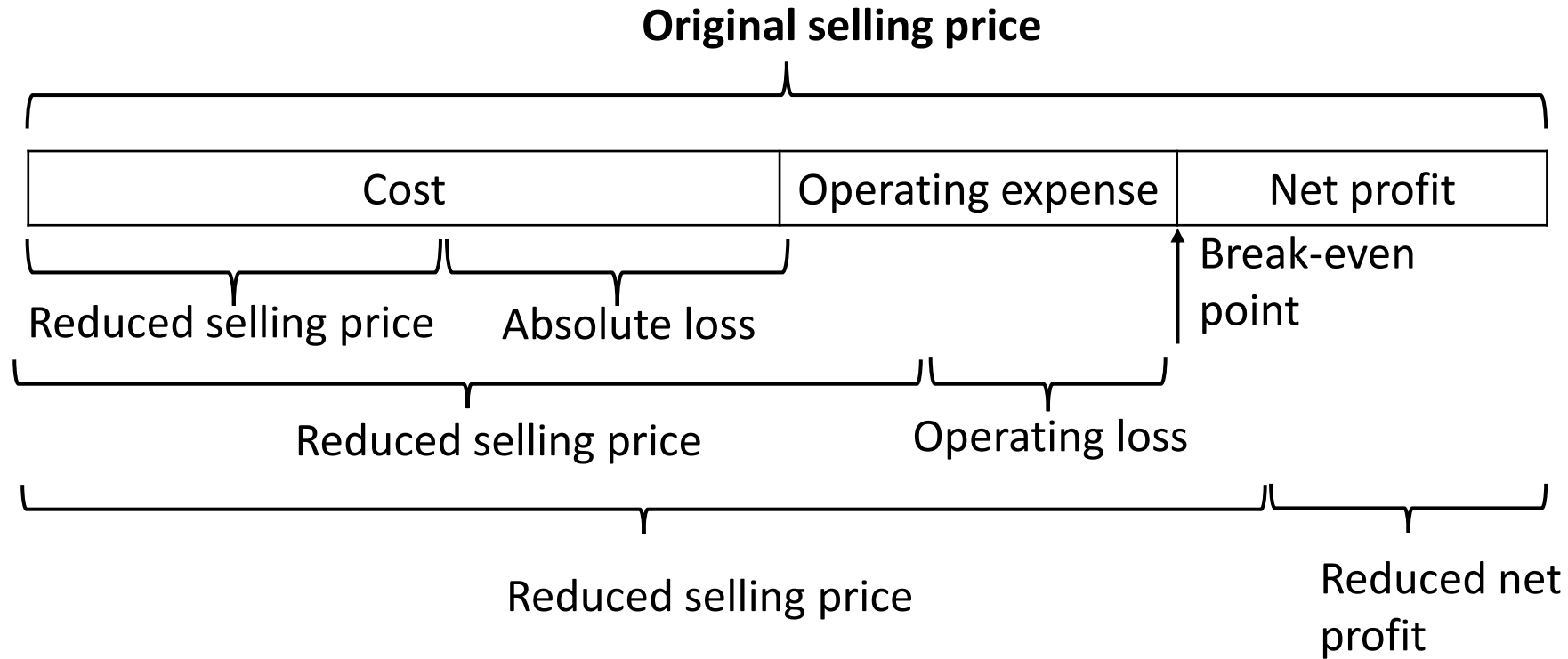
- Occurs when the reduced selling price of an item is below the break-even point but above the actual cost of the item.

4. Absolute loss or gross loss

- Occurs when the reduced selling price is less than the actual cost paid for the item. For example, buy an item for K30 sells it for K20.



- The diagram below illustrates the meaning of the above defined terms



- We can represent and understand the above information using formulas

i. Break – Even Point = Cost + Operating Expenses

ii. Operating Loss = Break – Even Point – Reduced Selling Price

iii. Absolute Loss = Cost – Reduced Selling Price

Example

Assume game stores paid K1600 for 52 – inch LCD Flat panel HDTV. If operating expenses are 30% of cost and the television is sold at K2000, find the amount of profit or loss, and show the information in the diagram.



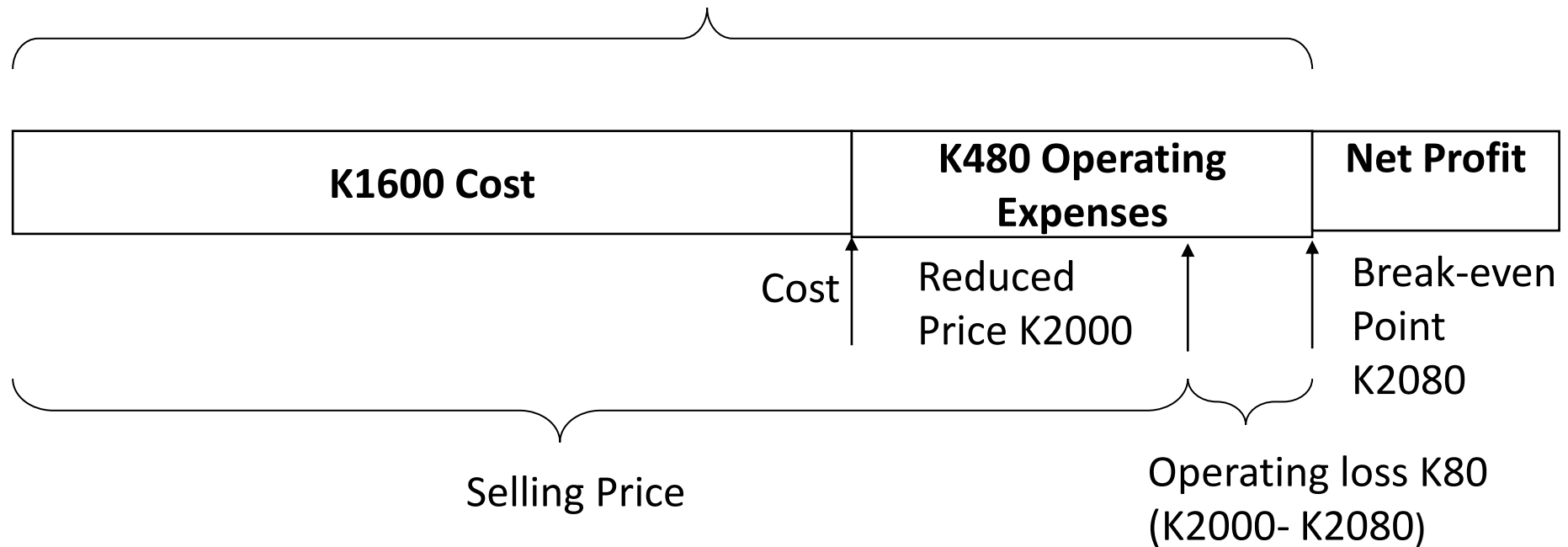
Solution

- Operating expenses = 30% of cost = $0.3 \times 1600 = 480$
- The break-even point is = cost + operating expenses = $1600 + 480 = 2080$
- Therefore, 2080 is the break-even point
- Therefore, Game stores makes a profit if it sells the television for more than the break-even point.
- Net profit/loss = selling price – break-even point = $2000 - 2080 = -80$
- Therefore, Game Stores makes a loss of K80



Diagrammatic representation

$$K1600 + K480 = K2080 \text{ Break – even point}$$



In-class exercise

A game table normally selling for K360 at Dick's sporting goods is marked down 30%. If the cost of the game table is K260 and the operating expenses are 20% of the cost. Find

- a) The operating loss
- b) The absolute loss
- c) Show the information in the diagram



Solution

a) The break-even point=cost + operating expenses

$$= K260 + 20\% \times K260 = K312.$$

$$\text{Reduced price} = K360 - (0.3 \times 360) = 360 - 108 = K252$$

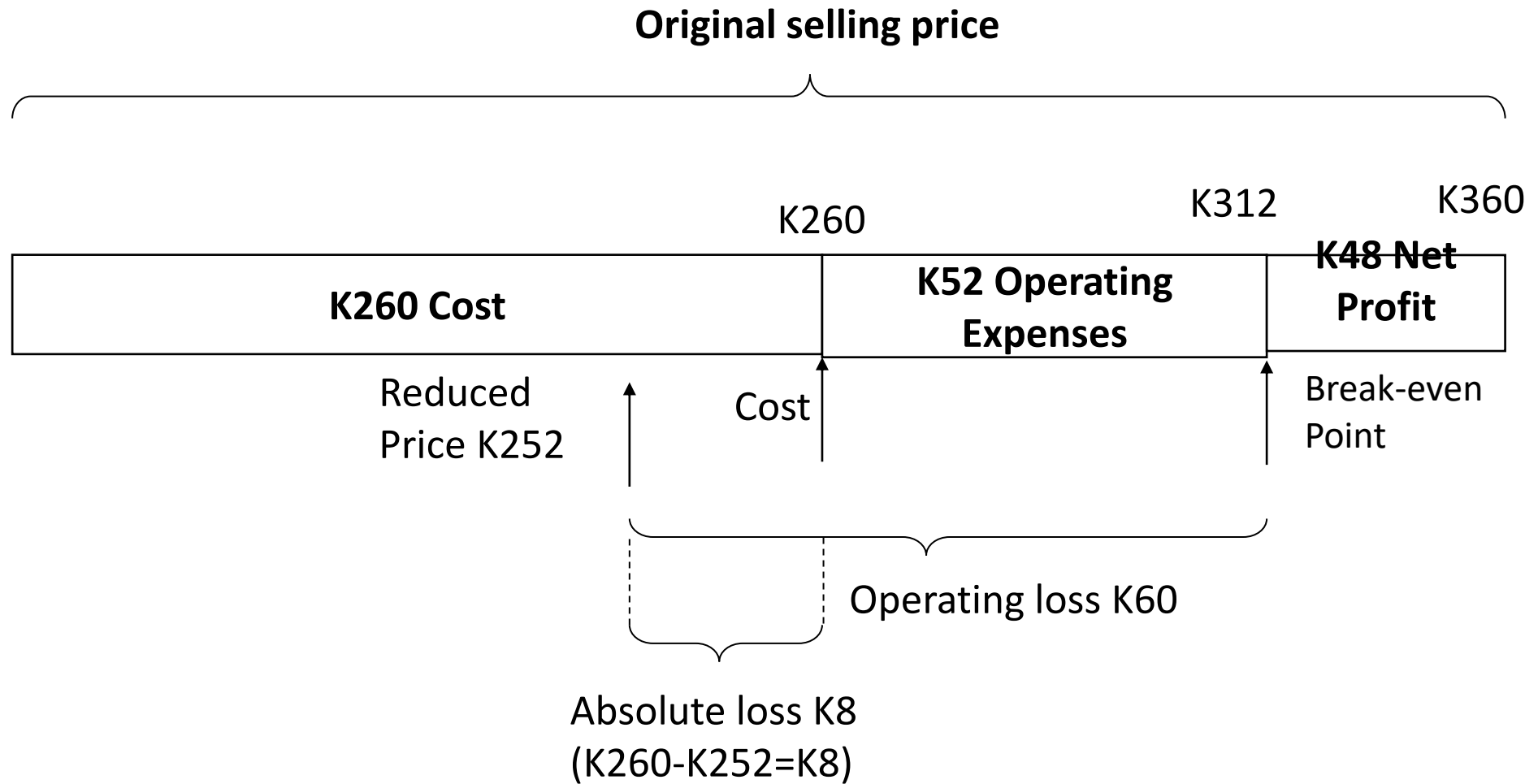
$$\text{Operating loss} = 312(\text{Break} - \text{even point}) - 252(\text{Reduced price}) = K60$$

b) The absolute or gross loss is the difference between the cost and the reduced price.

$$= 260(\text{Cost}) - 252(\text{Reduced price}) = K8$$



C) Diagrammatic representation



End of Unit 1

